Practice workbook · Calculating ratios

Writing and simplifying a ratio between two groups or a group and the whole.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: PSY-RM-13

1. Question 1 · part d · 2 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part d only.

1 Thomas recruited participants to investigate obedience to authority. The participants were asked to attend a university.

They were allocated to one of two conditions:

- Condition A: 30 participants were instructed to answer general knowledge questions in silence for 40 minutes by a man dressed in a uniform.
- Condition B: 30 participants were instructed to answer general knowledge questions in silence for 40 minutes by a man dressed in jeans and a t-shirt.

Thomas recorded how many participants obeyed the instruction to remain silent.

(a) Identify the dependent variable (DV) in Thomas' investigation.

(1)

- ☒ **A** Whether the man was wearing a uniform or jeans and a t-shirt.
- ☒ **B** The amount of time participants were given to answer the questions.
- ☒ **C** The number of participants who obeyed the man's instruction to remain silent.
- ☒ **D** The number of general knowledge questions participants were instructed to answer.

(b) Describe **one** participant variable that Thomas may have considered when conducting his investigation.

(2)

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1. Question 1 · part d · 2 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part d only.

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The results of Thomas' investigation are shown in **Table 1**.

	Obeyed instruction (remained silent)	Disobeyed instruction (did not remain silent)
Condition A: Man wearing uniform	22	8
Condition B: Man wearing jeans and t-shirt	11	19

Table 1

(c) Calculate the percentage of participants who **obeyed** the instruction, out of all the participants in the investigation.

(2)

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Percentage .....

(d) Calculate the ratio of participants who **disobeyed** the instruction, to all participants taking part in the investigation.

(2)

Ratio .....

(e) Thomas concludes that the clothes worn by an authority figure influence levels of obedience.

Give **one** way Thomas' results could support this conclusion.

(1)

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(Total for Question 1 = 8 marks)

1. Question 1 · part d · 2 marks · Working and answer

Working and answer

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2. Question 3 · part b · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part b only.

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3 Vanessa investigated age differences in driving speed. She conducted an observation by a road in the local town. Vanessa recorded the speed that young drivers and mature drivers drove their cars in miles per hour (mph).

The number of young drivers and mature drivers who were recorded driving at each speed is shown in **Table 3**.

	1mph to 10mph	11mph to 20mph	21mph to 30mph	31mph to 40mph
Young driver	1	4	9	12
Mature driver	1	5	14	18

Table 3

(a) Calculate the difference between the number of young drivers who were recorded driving at 1mph to 20mph and the number of young drivers who were recorded driving at 21mph to 30mph.

(1)

Difference .....

(b) Calculate the ratio of young drivers to mature drivers that Vanessa recorded driving at 31mph to 40mph.

(1)

Ratio .....

(c) Explain **one** improvement that Vanessa could make to her investigation.

(2)

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(Total for Question 3 = 4 marks)

2. Question 3 · part b · 1 mark · Working and answer

Working and answer

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3. Question 3 · part d · 1 mark

2020 NOV · Paper 02

Further practice · Attempt part d only.

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3 Becca wants to find out if there are sex differences in male and female brain functioning. She decides to use a laboratory experiment.

(a) State **one** feature of a 'laboratory experiment'.

(1)

(b) Describe how Becca could use volunteer sampling to gather the participants for her laboratory experiment.

(2)

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(c) Becca will give each participant a coordination task and a word-meaning task to test for sex differences in brain functioning. She decides to use a single-blind technique.

Describe **one** reason why Becca used a single-blind technique for her experiment.

(2)

(d) Becca conducted her laboratory experiment with 30 male and 25 female participants.

Calculate the ratio of male to female participants in Becca's sample.

(1)

Ratio .....

3. Question 3 · part d · 1 mark

2020 NOV · Paper 02

Further practice · Attempt part d only.

(e) Becca's results are shown in **Table 2**.

	Mean time taken to accurately complete the coordination task (seconds)	Mean time taken to accurately complete the word-meaning task (seconds)
Males	57.24	63.51
Females	64.43	51.36

Table 2

Calculate the difference between the mean time (seconds) taken by **males** to accurately complete the coordination task and word-meaning task.

You must give your answer to two significant figures.

(2)

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Difference in mean time (seconds) taken by males .....

(f) Explain **one** conclusion that Becca could make from her results.

(2)

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(Total for Question 3 = 10 marks)

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3. Question 3 · part d · 1 mark · Working and answer

Working and answer

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4. Question 2 · part c.i · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part c.i only.

2 Yolanda is conducting an observation of children's play in an early years day care setting. She observed children in the setting for four hours on a Monday.

(a) Define what is meant by an 'observation'.

(1)

(b) Describe how Yolanda could have used an opportunity sampling technique to select the children for her observation.

(2)

(c) Yolanda recorded the number of children who played with the different types of toys in the day care setting.

There were a total of 74 children recorded:

- 23 of the total children played with vehicles
- 11 of the total children played with dolls
- 29 of the total children played with role-play toys.

The remaining children played with arts and crafts toys.

(i) Calculate the ratio of children who played with dolls to those who played with arts and crafts toys.

(2)

Ratio .....

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4. Question 2 · part c.i · 2 marks

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Further practice · Attempt part c.i only.

(ii) Calculate the percentage of children who played with role-play toys as a percentage of the total number of children recorded.

You must give your answer to two decimal places.

(2)

Percentage .....

(d) Yolanda recorded the gender of the children who played with vehicles.

Her results are shown in **Table 1**.

	Number of children who played with vehicles
Males	18
Females	5

Table 1

Explain **one** conclusion that Yolanda could make about children's play from the results in **Table 1**.

(2)

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(Total for Question 2 = 9 marks)

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4. Question 2 · part c.i · 2 marks · Working and answer

Working and answer

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5. Question 1 · part c · 2 marks

2022 JUN · Paper 02

Further practice · Attempt part c only.

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1 Andrew conducted a laboratory experiment to investigate the capacity of short-term memory. He showed participants 20 trigrams and asked them to recall as many of the trigrams as they could.

(a) Andrew counted how many trigrams each participant recalled correctly.

Identify the type of data gathered by Andrew.

(1)

- ☒ **A** Qualitative data
- ☒ **B** Subjective data
- ☒ **C** Quantitative data
- ☒ **D** Secondary data

(b) Andrew used a standardised procedure when showing the trigrams to his participants.

Give **two** ways that Andrew could have standardised his procedure.

(2)

1 .....

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2 .....

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5. Question 1 · part c · 2 marks

2022 JUN · Paper 02

Further practice · Attempt part c only.

(c) Andrew sampled a total of 20 participants. A quarter of the participants were male.

Calculate the ratio of males to females in the participant sample used by Andrew in his experiment.

(2)

Ratio .....

(d) The number of trigrams recalled by participants correctly in Andrew's experiment are shown in **Table 1**.

	Number of trigrams recalled correctly				
	Between 1 and 4 trigrams	**Between 5 and 8 trigrams**	**Between 9 and 12 trigrams**	**Between 13 and 16 trigrams**	**Between 17 and 20 trigrams**
Males	0	2	3	0	0
Females	1	5	8	1	0

Table 1

Calculate the percentage of participants who recalled between 9 and 12 trigrams correctly, out of all the participants in the experiment.

(2)

Percentage .....

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5. Question 1 · part c · 2 marks · Working and answer

Working and answer

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6. Question 1 · part e · 2 marks

2023 JUN · Paper 02

Further practice · Attempt part e only.

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1 Gary conducted an experiment to test whether there was a difference in the ability of boys and girls to conserve number. He sampled 50 boys and 50 girls aged six years old.

Gary showed them the image in **Figure 1**.

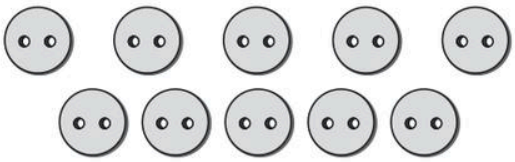


Figure 1

Gary asked each child to say whether the top row of buttons contained more, less, or the same number of buttons as the bottom row.

(a) Identify the independent variable in Gary's experiment.

(1)

- ☒ **A** age
- ☒ **B** gender
- ☒ **C** ability
- ☒ **D** conservation

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(b) Give **two** ways that Gary could ensure his experiment is ethical.

(2)

1 .....

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2 .....

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6. Question 1 · part e · 2 marks

2023 JUN · Paper 02

Further practice · Attempt part e only.

The results of Gary's experiment are shown in **Table 1**.

	Number of children who said the top row contained less buttons than the bottom row	Number of children who said the top row contained the same number of buttons as the bottom row	Number of children who said the top row contained more buttons than the bottom row
Boys	5	13	32
Girls	1	25	24

Table 1

(c) Calculate the percentage of girls who said that the top row contained the **same** number of buttons as the bottom row, out of all the children in the experiment.

(2)

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Percentage .....

(d) Calculate the range for the number of children who said the top row contained **more** buttons than the bottom row.

(1)

Range .....

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6. Question 1 · part e · 2 marks

2023 JUN · Paper 02

Further practice · Attempt part e only.

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(e) Calculate the ratio for the number of children who said that the top row contained **more** buttons than the bottom row, to all children taking part in the experiment.

(2)

Ratio .....

(f) Explain **one** conclusion that Gary could make about the ability to conserve number between boys and girls from the results in **Table 1**.

(2)

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(Total for Question 1 = 10 marks)

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Working and answer

7. Question 3 · part b · 2 marks

2024 JUN · Paper 02

Further practice · Attempt part b only.

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3 Savannah is investigating whether there is a relationship between listening to music and happiness. She asked participants to record the number of hours they spent listening to music in a week. They also scored their level of happiness on a scale of 1 (being very unhappy) to 10 (being very happy).

Savannah sampled a total of 180 participants aged between 18 years old and 65 years old. A third of her sample were aged between 18 years old and 25 years old.

(a) Calculate the range for the age of all participants.

(1)

Range .....

(b) Calculate the ratio of participants aged between 18 years old and 25 years old to the remaining participants in the sample used by Savannah.

(2)

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Ratio .....

(c) Savannah found that those listening to less than one hour of music in a week scored themselves at 2 or below on the happiness scale. Happiness scores increased for those who listened to more music, with those who listened to over 12 hours of music in a week usually scoring 9 or above.

Explain **one** conclusion that Savannah could make from her findings about the relationship between music and happiness.

(2)

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(Total for Question 3 = 5 marks)

7. Question 3 · part b · 2 marks · Working and answer

Working and answer

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